

Homework (Jalapeno)

- Q1.** The ancient Egyptians used $|x - 5|$ to measure the distance from a temple. If the distance is less than 3, what is the solution set in interval notation?
- (A) $(2, 8)$
 (B) $[2, 8]$
 (C) $(-\infty, 2] \cup [8, \infty)$
 (D) $[2, 8)$
 (E) $(2, 8]$
- Q2.** The Great Pyramid of Giza's height above sea level is modeled by $|h - 150| \leq 20$. What is the range of possible heights h ?
- (A) $130 \leq h \leq 170$
 (B) $130 < h < 170$
 (C) $130 \leq h < 170$
 (D) $130 < h \leq 170$
 (E) $120 \leq h \leq 180$
- Q3.** The Phoenicians used $|x + 2| > 5$ to navigate the Mediterranean. What is the solution set on a number line?
- (A) $(-\infty, -7) \cup (3, \infty)$
 (B) $(-7, 3)$
 (C) $(-\infty, -7] \cup [3, \infty)$
 (D) $(-7, 3)$
 (E) $(-7, 3]$
- Q4.** The Rosetta Stone's age is estimated using $|a - 2000| \geq 500$. What is the solution set for a ?
- (A) $a \leq 1500$ or $a \geq 2500$
 (B) $1500 \leq a \leq 2500$
 (C) $a \leq 1500$ and $a \geq 2500$
 (D) $1500 < a < 2500$
- (E) $1500 < a \leq 2500$
- Q5.** Ancient Babylonians measured the distance between cities using $|d - 50| < 20$. What is the solution set for d ?
- (A) $30 < d < 70$
 (B) $30 \leq d \leq 70$
 (C) $30 \leq d < 70$
 (D) $30 < d \leq 70$
 (E) $20 \leq d \leq 80$
- Q6.** The ancient Greek Parthenon's length is estimated using $|l - 70| \leq 10$. What is the solution set for l ?
- (A) $60 \leq l \leq 80$
 (B) $60 < l < 80$
 (C) $60 \leq l < 80$
 (D) $60 < l \leq 80$
 (E) $50 \leq l \leq 90$
- Q7.** The Terracotta Army's height above ground level is modeled by $|h - 5| > 2$. What is the solution set for h ?
- (A) $h < 3$ or $h > 7$
 (B) $3 \leq h \leq 7$
 (C) $3 < h < 7$
 (D) $h \leq 3$ or $h \geq 7$
 (E) $3 < h \leq 7$
- Q8.** The Great Wall of China's length is estimated using $|l - 4000| \geq 1000$. What is the solution set for l ?
- (A) $l \leq 3000$ or $l \geq 5000$
 (B) $3000 \leq l \leq 5000$
 (C) $3000 < l < 5000$
 (D) $3000 \leq l < 5000$
 (E) $3000 < l \leq 5000$

Open Ended Questions (FRQ)

- Q9.** In ancient Rome, the distance between two landmarks was estimated to be $|d - 20| < 5$. Find the solution set for d and graph it on a number line.
- Q10.** The Aztec calendar's age is estimated using $|a - 500| > 200$. Find the solution set for a and express it in interval notation.

Chef's Tips

- Tip 1: Ensure students understand the concept of absolute value inequalities.
- Tip 2: Emphasize the importance of expressing solutions in interval notation.
- Tip 3: Encourage students to visualize the solutions on a number line.